

HEURISTIC ABOUT GEOMETRIC POSITIONING AND APPLICATIONS

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WHO AM I?

I'm AIMER7, a former arena FPS player, who was also a top Tracer and Soldier in early Overwatch (from season 1 to season 6). I'm now mostly playing KovaaK's FPS aim trainer, and even published an aim workout guide for it. It seems that people enjoy my free guides¹ (and my private coaching too, for people who are interested), so I decided to make a new one about what I call geometric positioning. Hope you guys enjoy it.

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¹it's a lot of work, but as a former scientist too, I know how important free access to knowledge is. I therefore won't make people pay for this kind of stuff. If you guys judge that this work deserves some money, feel free to donate what you want on my twitch page

1. INTRODUCTION

In every kind of game, from MOBA to FPS, the concept of positioning is used and abused by people to justify why something worked or failed, but no distinction is made between its different aspects. Good players generally have decent intuition about what is a good or a bad position, but this intuition is phenomenological² and completely brute-forced³. Usually, not to say always, if you ask someone to judge positions in a concrete situation, his only answers are going to be “good” or “bad” and no heuristic is ever given to explain **why** it is so. This inability to abstract and to consciously express one’s own intuition is very limiting:

- some aspects of positioning are universal and **should** translate from game to game⁴. Staying intuitive makes people conflate these universal aspects with the game-specific ones,
- by changing maps, even though the effective **geometry** is preserved, a player have to rebuild an intuition for something he actually already knows,
- intuition is very hard (if not impossible) to share. Convincing yourself (or others) that you did a mistake is therefore very hard,
- in a real game situation where you have time to think about positioning, staying at the level of intuition might be unfruitful if the latter stays silent. Having a heuristic allows you to “make it talk” while playing (at least, if you allocate some brain power to it), which can fix your mistakes or make you discover new ways to abuse positioning live,
- having a heuristic allows you to “make it talk” outside of the game too. You can as a result discover and learn (potentially) better positions waiting to be applied (or tested) in real games.

In this guide, I decide to split positioning skills in two classes. To understand that, suppose that you are playing a shooter coming with all of its *game-specific details* (classes, skills, weapons, ...). Your goal is of course to win the game. Now, even though the winning conditions are very different from shooters to shooters (say, death-match vs objective based games), there always exists strategies with better win-rates. These strategies are part of the *game-sense*. They are global things, as in, they are heavily influenced by game-specific details, and involve a significant part of the map for a rather long time-scale⁵, say, of the order of 2 to 20 seconds (sometimes even more in the Battle Royal genre). In particular, they are taking into account players *positions*⁶ at one time and tell them how to reposition themselves at the next⁷. These positions with long time-scale and a strategic purpose are part of the *game-sense positioning* skills. Let me give some examples of the latter:

- in team games, playing properly as a team by knowing when to rotate, when to engage, when to disengage, or when to position yourself with respect to your mates⁸,
- in death-match games, knowing what are the “hot spots” on the maps to meet and to frag more people, or the routes that maximize your damage output,
- in duel games, knowing the routes, the places not to commit that would eventually lead to your death, or the ones to gather information,
- ...

Now, shooters being shooters, you necessarily have to kill people (or avoid being killed) at one point or another. Fights are local things: they generally happen between two players, on some precise place of the map, and for much shorter time-scales. In what follows, the time-scale of fights is considered to be of the order of 300 milliseconds. Contrary to game-sense

²based on experience

³it needs a fuck ton of trials and errors

⁴of the same genre obviously

⁵of course, this time-scale is not absolute and depends of situations

⁶the orientation of the players, that is where they are actually looking at, is always taken into account

⁷if the first time is called t and the timescale is called τ , then “at the next time” refers to the time $t + \tau$, so the difference of time between the two instants can be quite big

⁸notice that it is *when* that matters in this case, not *how*

positioning skills whose purpose is to win the game, the purpose of *mechanical positioning* is to maximize damage output while minimizing damage taken during fights. These two last goals are generally opposing one another (it would be too simple otherwise, right?). Let me give some examples of mechanical positioning skills:

- crosshair placement,
- knowing when it is more important to maximize damage output instead of minimizing damage taken. The other way around too, or generally, to find a good balance between the two,
- targets selection, as in, knowing who to attack or/and who to dodge (they are not necessarily the same enemy),
- dodging skills,
- how to abuse the map geometry to hit “for free” or to avoid being hit “for free”. Let’s name it *geometric positioning*.
- ...

Geometric positioning is universal and is what this guide focuses on. The first point is universal too but I suppose you guys are good at it, and it’s generally hard to say anything abstractly about it anyway. The second point can only be judged on a case by case basis. Same for the third point, which is also heavily influenced by game-specific details. Dodging skills are universals, but they are influenced by the game ground-speed and ground-acceleration. They are very advanced, and you first need to understand geometric positioning to become good at them “quickly”. This guide would be way too long if I were to talk about them in details, they are consequently omitted.

So far, I’ve separated positioning skills in two classes: objective-driven (game-sense positioning) and combat-driven (mechanical positioning) ones. Now, these two classes are clearly not independent⁹, but let me insist that **the game-sense positioning skills always take precedence over the mechanical ones**. Your **primary goal** is to win the game. Being the most efficient in fights is only a **secondary goal**. As a result, you should always consider your mechanical positioning to be constrained and influenced by the game-sense one. In particular, there exists a seemingly trivial yet very strong criterion to judge if a decision to take a fight (and the way to take it) is good or bad.

The aggressive aimbot criterion: *a decision to take a fight (or a way to take it) is bad if you have no way to win it even by assuming you have an aimbot.*

The passive aimbot criterion: *a decision to avoid a fight (or a way to avoid it) is bad if you have no way to lose it even by assuming your enemy has an aimbot.*

This criterion is not as trivial as it looks and is very useful in team games. As an example, suppose that you want to fight two players: can you out-damage them assuming you aimbot? More concretely, assume that you have 157 hp and deal 100 damage per second thanks to the aimbot. The players you face have 100 hp and 50 hp respectively, and each of them (realistically) deals 80 damage per second. Do you take the fight in this case? With good target selection, you need 0.5 second to kill the 50 hp target first. Assuming instant target switch, one more second is needed to kill the 100 hp target left. Yet, you get hit for 80 hp during the first 0.5 second, and now have 77 hp against a 100 hp target. You cannot win the fight in these conditions, even though you have more hp and a better DPS. The decision to take the fight is therefore bad (applying the aggressive aimbot criterion). If the enemy applies the passive aimbot criterion, they conclude that there is no way to lose the fight by sticking together. You should hence expect them to rush you. If they don’t, you can always think of a smarter way to engage the fight.

Enough useless talk. The first part is rather long and deals with the theory of geometric positioning. The terminology defined is necessary in order to state a heuristic usable in real game situations to improve your combat skills. Applications are presented as exercises in the second part. The reader is invited to think seriously about these exercises, and to look at their solutions in the third part to check for his answers rather than once stuck for 30 seconds.

⁹if you fight like a god, you better use your combat skills and play more aggressively

To conclude this introduction, let me point that this guide won't benefit everyone to the same degree. Good players of slow games with very low time to kill, and a lot of indoor actions like Counter-Strike or Rainbow Six: Siege, most likely know all of the content of this guide already, at least intuitively. Indeed, in these games, the raw mechanics side of aiming skills matter much less than their game-sense and geometric aspects. In fast games with more movement, higher time to kill (more time to move and to reposition quickly while fighting), and more outdoor actions like Quake, Overwatch, Fortnite, Team Fortress 2 and Apex Legends (say), a lack of geometric positioning skills is not that detrimental to win games, because it can easily be compensated by raw aiming skills and other game-sense skills¹⁰. At high skill though, understanding the content of this guide can make a huge difference by giving you an edge over your enemies. This guide might therefore benefit the second class of players much more than the first. By making the intuition about geometric positioning explicit, top players of Counter-Strike and Rainbow Six: Siege might still learn one thing or two, and avoid making dumb mistakes (or maybe not, because this guide emphasizes dynamical fights where repositioning yourself is possible. In these slow games, the fights are static, and avoiding getting hit by protecting yourself behind corners is more important than trying to engage because of the very low t.t.k.).

2. AN EFFECTIVE HEURISTIC

The goal of this part is to present a very simple and effective heuristic to increase your combat skills (both offensive and defensive). The heuristic is presented in the easiest possible situation¹¹. More complexity is added in the second part.

2.1. 2-dimensional and 3-dimensional aim. Shooters are taking place in a 3-dimensional environment (a map), but the perspective is first person, which means that it's closer to be 2-dimensional. More precisely, as a player, your perspective is a sphere minus one or two points (the north and south pole), and this is what makes vertical angles hard: your sensitivity to do a full turn at the equator is the same at every latitude, which means that the closer to the poles you aim at, the slower your horizontal sensitivity **feels**. Now, notice that if you're looking at an object in real life, say, a chair or a pen on your desk, the 3-dimensional space (in particular the depth) is easily reconstructed by your brain. This is possible because you have two eyes that are a bit far apart. They therefore see the same object with a different angle which gives this impression of depth.

A common exercise to convince yourself of that is to take a pen in one hand and its cap on the other. Now, obviously, if you try to put the cap back on the pen with both eyes open, it's very easy. Try to do it with only one eye opened, and notice that it becomes much harder. In video games, you only have one 2-dimensional image at once on your monitor so the 3-dimensional perspective is kind of "fake", like the cube below.

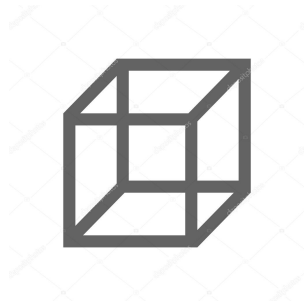


FIGURE 1. Do you see a cube from above, from below, or just a collection of 2-dimensional lines?

¹⁰Said differently, you don't need a lot of raw aim to be a top Counter-Strike or Rainbow Six: Siege player but you do need a ton of geometric positioning skills. Now, the opposite is true for Quake (especially the team modes like 2v2 or CTF), Overwatch, Fortnite, and Apex Legends

¹¹spreadless, simple hitbox, isotropic game

In this image, most people will see a cube from above or from below, and it is easy to change interpretation at will (try it). Yet, the image is 2-dimensional, so why do you want so hard to see a cube? Can you force yourself to see a simple collection of 2-dimensional lines? If yes, then you have to understand that all of these interpretations of the “cube” image above are legitimate: nobody is right or wrong, but depending on the context, one of the three interpretations might be of better use. Notice that forcing yourself to see a 2-dimensional depth-free collection of lines is less natural and harder too. The same is true for the two following aiming-style:

- **3-dimensional aim:** this is what you do naturally. You aim at models evolving in three dimensions, so you have an impression of depth and therefore move your crosshair accordingly. This is a good aiming-style to predict movements but it makes some easy shots harder and is actually making you believe that two equivalent situations are different. For example, suppose that there is a rock 3 meters in front of you a bit on the left of your crosshair, and a guy crouching behind it. As he uncrouches, you see his head and can therefore aim at him. Now, suppose that you’re in some big plain without obstacles. A target of the same apparent size as the head before appears 100 meters away from you (so it feels) at the exact same place on your monitor. Your brain will interpret both situations differently, which might fuck your aim up, even though the coordinates and the size of the target on your monitor are exactly the same.
- **2-dimensional aim** or “projective” aim: as with the cube example above, this is what you do when you force yourself to see a 2-dimensional image (hence trying hard to remove the depth). This is not natural and requires some significant brain power. You therefore can’t do that at all times (and shouldn’t anyway). This is very good for pre-shooting, when crossing corners (you try to aim at their very edge while moving), or while flicking. In particular, it makes predicting movements harder because when you 2d-aim, you actually aim at a point in (a 2d) space rather than at a model. When you stand still in an aim trainer and play 1wall6targets kind of scenarios, it simulates 2d-aim. If you’re very good at those, you should be very good at 2d aim in real games too. This aiming-style is therefore more “geometric” than 3d-aim, and can be abused in many situations. It is however not reactive at all, because you completely ignore the intuition about movement and try to only keep the spatial component of aim.

2.2. Player-dome and all that. To begin, let me recall that each shooter allows you to move in eight directions (forward, backward, left, right, and the four diagonals), plus jump and crouch. Depending on the physics of the game, the eight directions are not necessarily equivalent, that is, some directions might be privileged or act differently than others. In the simplest case, where the ground-acceleration and the ground-speed is the same for all directions, moving for a short period of time in one direction or another will make you travel the exact same distance. In particular, by “tapping” movement keys accordingly, you can actually move in more than eight directions. Recalling that the time-scale of fights is of the order of 300 milliseconds, the possible movement of a player during this time is a disk of radius $300ms \times v$, where v is the ground speed of the game. This disk is called the *player-disk*: it is centered on the player at some time, and indicates all accessible positions 300ms later. Note that when you fight a player, you aim at him and he aims at you, so the orientation is (relatively) left unchanged. Hence why you have to imagine that the orientation is fixed on figure 2.

If the game has lower strafe-speed than forward and backward speed, then the boundary of the player-disk is not a circle but an ellipse. If the backward speed is different than the forward speed, then it’s something more complicated. Still, the terminology I’m going to use is player-disk even for non-isotropic games¹².

Now, a player can jump and the space he moves in is 3-dimensional. The analog of the player-disk in this case is the *player-dome*. This is a volume and not a surface, see figure 3.

¹²where the ground-speed or the ground-acceleration is not necessarily the same for all directions

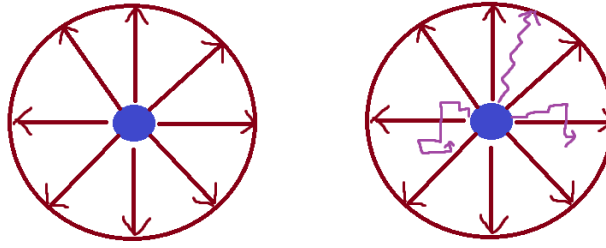


FIGURE 2. Player-disk seen from above. The player is in blue and does not change orientation (so he doesn't move his mouse), and the eight movement directions are indicated in red. On the right picture, the purple directions are possible by tapping movement keys. Any point of the disk is accessible by the player in a time lapse of 300ms

The player-dome quantifies the accessible space of the player for a time lapse of 300ms. That

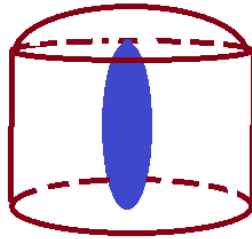


FIGURE 3. The player-dome is the accessible volume of a player during a time lapse of 300ms

is, it quantifies the movement *freedom* of a player. In particular, as a player, you also have a player-dome attached to you. When the player-dome does not intersect any obstacles (like a wall, a corner, a door, or anything really) it is said to be *free*. The player-dome on figure 3 is free. What happens to the player-dome when it meets obstacles? For example, when a player is close to a wall? If there is no way for the player to cross the wall (depending on game-specific details, it might be possible), then the player-dome does not extend through it. For example, if the back of a player touches the wall as in figure 4, then the volume of the player-dome is half of the free case. This shows that the possible movement of a player is indeed quantified nicely by the volume of the player-dome, for a short time-scale at least. There is however some subtleties to take into account. Suppose that you're fighting a player at the **same height**¹³, and your aim is good enough so that your orientation is always toward him (as in, you don't miss blatantly and your crosshair is always very close to the enemy model) and reciprocally,

- at very close range, your own model is an obstacle for the enemy player-dome, unless the game allows you to go through each other¹⁴,
- at mid and long range, for a short time-scale, the difference of apparent hitbox size (of your enemy) between a forward-dominant or backward-dominant motion is neglectable. In particular, by 2d-aiming, the collection of all relative motions¹⁵ between backward-dominant and forward-dominant movement (made by your enemy and thus seen from your perspective) are identical,

¹³so you have an horizontal aim duel

¹⁴some subtleties can happen due to the netcode, but let's forget about them right now

¹⁵every kind of trajectory that could happen on your monitor if the enemy moves during the 300ms

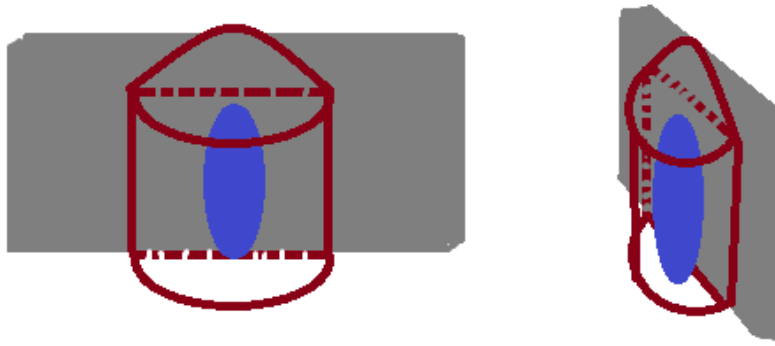


FIGURE 4. The player, in blue, is touching the wall, in grey. The player-dome volume is half of the free case

- a corollary of the previous point is that by 2d-aiming, you have a **mirror-symmetry** between forward-dominant motion and backward-dominant motion in this horizontal case. That is, aiming at the two players on figure 5 is equivalent to the free case of figure 3 (for short time-scales at least),
- a reformulation of the above corollary is to say that, in a mid to long range horizontal fight, you only have to correct your aim horizontally. This seems trivial, but it has a lot of consequences. One of them that I won't discuss in this guide is that good dodging skills should use all eight directions (+forward only, +backward only, and diagonals too). Indeed, by doing so, you add variability in your movement seen from your opponent's perspective, and this is obviously much more effective than plain left-right,
- remember that you also have a player-dome too (let's call it the *self-dome*), and the mirror-symmetry holds true for your opponent about you too!
- **advanced remark:** in real games, model animations are sometimes changing between a forward and a backward movement. Doing some triangle-dodges that alternates relatively quickly, say, "forward + left strafe" with "backward + left strafe" can actually fuck the aim of the opponent. In any case, if you're close enough to your opponent, and even when the ground-acceleration is finite or low, the mirror-symmetry does not hold true because the difference of apparent hitbox size and relative motion become noticeable.

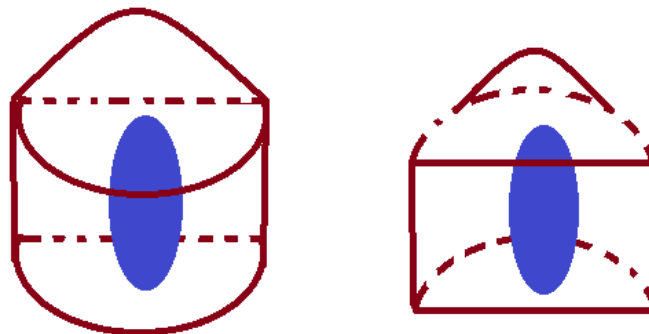


FIGURE 5. The left picture shows a forward-dominant motion, while the right one shows a backward-dominant motion (during 300ms). From your perspective, by 2d-aim, and by ignoring change of hitbox size, the two situations are equivalent. In particular, they are equivalent with the free player-dome case too.

What happens now, if there is some verticality? Say, your enemy is high ground and you're low ground. The mirror-symmetry is not necessarily true for many reasons (models being non-cylindrical, small head hitbox, significant change of the visible part of the hitbox, ...) but the main one is that you often have to adapt the pitch (of your aim) too. I'll treat many of these cases later, so it's no need to enter into too much details right here.

Notice that if we take projective aim seriously, then the **volume** of the player-dome is **irrelevant** because what you see on your monitor is a picture in 2 dimensions. What *should* matter is the **apparent surface of the player-dome from your perspective**. To convince you of that, let's do some simple geometry with the example of figure 4. By the mirror-

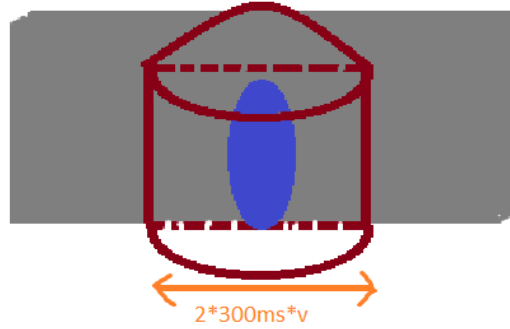


FIGURE 6. You're looking straight at the enemy whose back touches the wall. Here, the width of the player-dome is its diameter, that is, $2 \times 300\text{ms} \times v$

symmetry above, only the horizontal motion of your enemy matters. If you're in front of your enemy, as in figure 6, then during the 300ms time-scale, your enemy has a lot of (apparent) space at his disposal. What happens now, if you look at him from the side as shown on figure 7? The apparent width of the player-dome is half of the straight case. There is therefore much less freedom of movement for your opponent, and you therefore have to aim less: you don't need to move your crosshair as much as in the straight case! Obviously, the apparent width

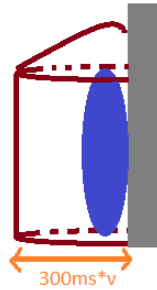


FIGURE 7. From the side, the width of the player-dome is only $300\text{ms} \times v$, that's half of the straight case

varies continuously with the angle from these two extremal cases. In order to make sense of that a bit more formally, let us define a new concept. Suppose that the player-dome of your enemy is non-free, as in, there is some obstacle reducing its apparent surface (be it a wall, a corner, a rock, the edge of a high-ground, ...). What happens is that, **at fixed distance**¹⁶ there exists a (non-necessarily unique) direction from your perspective that **maximizes** the apparent surface of the player-dome. Such a direction is called a *normal*. In the previous

¹⁶imagine yourself moving on a circle (or a sphere) around the player. You're forced to, because it is always possible to maximize the apparent surface of the player-dome by closing the distance. By doing the latter, you're not winning nor losing anything (at least for very short time-scale) because this is most often than not symmetric (you have a self-dome too, keep that in mind)

example of an enemy standing next to a wall, the normal is what mathematicians would call a normal (that is, a perpendicular line to the wall), hence the name. See figure 8 where the normal is represented in green. What about the normal of a free player-dome? In this case (at

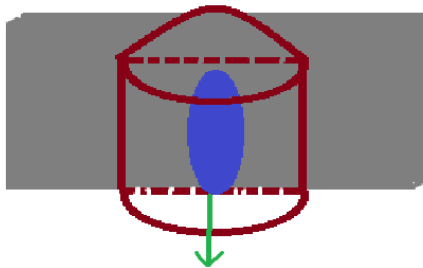


FIGURE 8. The normal, in green, is the direction that maximizes the apparent surface of the player-dome

fixed distance as always), any direction maximizes the apparent surface by definition, so any direction is a normal, as shown on figure 9. This means that there is no way to minimize the apparent surface because there is no obstacle. Of course, if from your perspective an obstacle

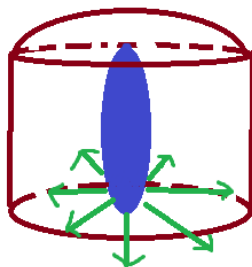


FIGURE 9. Any direction is a normal in the free case

exists between you and the enemy, and it reduces the apparent surface of the player-dome, it might be interesting to abuse it. Indeed, by 2d-aim, a smaller apparent surface of the player-dome means that the enemy has less space to move on your monitor¹⁷, thus making (3d) aim easier. This is not exactly true, because you might lose too much of the enemy model in the process, and it is **hard** to be precise and to read the movement of a very small hitbox (say, if you only see the head or the feet). Let us call *model-dome* the apparent surface of the enemy model in its player-dome seen from your perspective, and *empty-dome* its complementary (the apparent surface of the player-dome with no model), see figure 10. The (apparent surface of



FIGURE 10. The left picture shows the model-dome in orange, while the right picture shows the empty-dome in yellow

the) empty-dome is what quantifies exactly the movement freedom of your enemy (from your perspective, so as seen on your monitor), while the (apparent surface of the) model-dome is

¹⁷not necessarily for him

what you want to aim at to hit the enemy. Note that this is perfectly symmetric, because you also have an empty-dome and a model-dome. That is, for short time-scales, in order to hit more, you want to maximize the model-dome of your enemy and to minimize his empty-dome as much as you can. In order to get hit less, you want to minimize (the apparent surface of) your model-dome and to maximize (the apparent surface of) your empty-dome as much as you can from the enemy's perspective.

With this new terminology, notice that *taking an angle* with respect to the normal is reducing the empty-dome in the example of figure 4. The bigger the angle, the smaller the empty-dome. You don't want to take a too big angle, because by doing so, you might automatically allow your opponent to take an angle on you. Again, keep in mind that you have a player-dome too. As an example, think of a map with an infinite wall¹⁸ and a player touching it. In the straight case (so you look at the enemy in the direction of the normal, as in, no angle is taken), you don't win anything from the free case because of the mirror-symmetry. Now, in the extremal case of taking an angle of 90°, you're also touching the wall, hence the enemy took an angle on you too and the situation is perfectly symmetric: you didn't take any advantage over your opponent either. I can now state the following heuristic.

2.3. The heuristic about geometric positioning. This part is short, but if you understand the following heuristic with all of the terminology introduced in the previous section, then you should have no problem doing the exercises of application. For clarity, I'm making everything explicit, which makes it a bit hard to get but unambiguous.

Heuristic: *In a fight, a positioning is good if the apparent surface of the model-dome of your enemy (seen from your perspective) is bigger than the apparent surface of your model-dome (seen from the perspective of your enemy)¹⁹, and if the apparent surface of the empty-dome of your enemy (seen from your perspective) is smaller than the apparent surface of your empty-dome (seen from the perspective of your enemy)²⁰. A good **compromise** to achieve this is to **take an angle on** (one of the normal of) the enemy player-dome without allowing him to take an angle on (one normal of) your self-dome.*

This heuristic is limited by the following facts:

- it is almost never possible to maximize the model-dome and to minimize the empty-dome of your enemy simultaneously. This means that you have to make a choice depending on the situation and on your aiming skills. If you're very good at precision aim but not very reactive, you might want to minimize the empty-dome more than you maximize the model-dome of your enemy. If you're very reactive but not that precise, you might want to maximize the model-dome more than you minimize the empty-dome of your enemy. Note that this choice is sometimes enforced by the geometry of the situation, so you can't always act on it,
- the same is true for your own model-dome and empty-dome. You therefore have to find a balance that still gives you a positioning advantage. According to the heuristic, this balance is very often achieved by taking an angle,
- if your weapon has spread, you have to take it into account: even if you manage to reduce the empty-dome of your enemy to zero, but by doing so his model-dome becomes very tiny, spread might forbid you to land one shot even if your aiming skills allow you to,
- if there is head-shot multiplier, you have to take it into account too,
- if the enemy is very good, he might expect you to position yourself according to the heuristic in the situations he has an intuition for. For example, you can maximize your empty-dome and minimize your model-dome by hiding behind a rock, with only your head peeking. A smart enemy might expect you to do that and pre-shot above the rock. I'll give this situation as an example later in the guide,

¹⁸big enough really

¹⁹it makes him easier to hit than you are if we forget about movement

²⁰you have more space to move than he does, and hence can dodge more and have to (3d) aim less

- dodging skills add a lot of mind game to this: you expect your opponent not to reduce his empty-dome, but he can do it with good timing²¹ (say, if you're late on him and therefore miss). This becomes too advanced for this PDF,
- in a fight, the positions are updating very quickly, especially if there is dodge, so the normal is updating too, so what "taking an angle" means is updating too, especially if the geometry is not simple! This isn't a real limit because you can actually see it happening live and react to it.

This ends this theoretical part. In the next part, I'm going to apply this heuristic to concrete examples, and I'll try to refine it so you guys understand how game-specific details (like spread, hitbox size, etc.) can change quite some stuff.

3. EXERCISES OF APPLICATIONS

In this part, a lot of concrete examples are presented as simple exercises. The correction is given in the next section, but I highly recommend that you take some time to think seriously about these situations by using all of the above concepts and terminology. If you find different solutions than the ones in the correction, feel free to come to me, maybe your solution works too. If you disagree with a solution, come to me too, you might be right and there might be some game-specific detail changing a lot of stuff, and it therefore deserves to be added here to refine the understanding of this heuristic. In what follows, you should assume that you're using a tracking or semi-auto weapon. That is, the rate of fire is of the order of (at most) 300ms²². If the gun has a too slow rate of fire (like a sniper), then there is no dynamics so it is less interesting.

3.1. The Infinite Wall. Suppose that you fight an enemy whose back is touching an infinite wall as in the following figure,

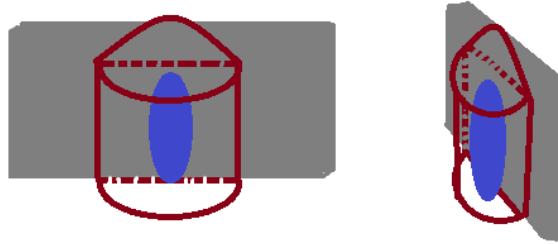


FIGURE 11. The infinite wall example

- (1) Draw the normal in both figures. Is it unique?
- (2) Is it more important to minimize the empty-dome or to maximize the model-dome here? Why?
- (3) What are the angles from the normal that minimize the empty-dome?
- (4) Would you gain any advantage by taking such an angle on your enemy? Why?
- (5) Give at least two optimal angles with respect to the normal that would favor you in a fight against this target.
- (6) Why is 45° not an optimal angle?

3.2. The Corner Case. Suppose that you fight a guy stuck in a corner (it happens when people camp in houses in BR, or when people are in their fort in Fortnite) as in the next figure,

- (1) Draw the normal. Is it unique?
- (2) What are the angles from the normal that minimize the empty-dome?
- (3) Would you gain any advantage by taking such an angle on your enemy? Why?
- (4) Give two optimal angles.

²¹think of any kind of wall-dodge

²²McCree's gun works even if the rate of fire is slower than 300ms

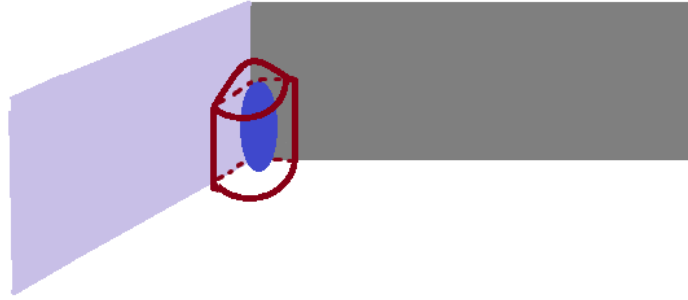


FIGURE 12. Your enemy is in the corner

- (5) Is the normal direction (the 0°) good in this case? Why?
- (6) What about 35° ?

3.3. Behind The Corner. Suppose that you're playing an **FPS** and there is no spread. An enemy is hiding behind a corner and can peek at any time, as in the next figure,

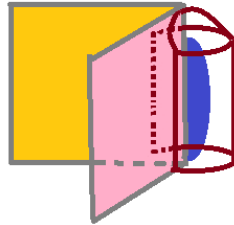


FIGURE 13. An enemy is hiding behind a corner and can peek at any time

- (1) If the enemy is standing close enough to the corner so that his player-dome extends around it, is the normal unique? Draw one.
- (2) If the enemy is standing behind the corner so that his model-dome is zero from your perspective, at what angle would you engage a fight around the corner? Where are you positioning your crosshair?
- (3) Did you minimize or maximize something in the previous question? What exactly? How to do that even more efficiently by playing with the distance to the corner?
- (4) If the guy is now peek-shooting, how are you reacting? What should you avoid to do?
- (5) Now suppose that you're the guy behind the corner and someone is trying to engage you. Can you think of a way to defend yourself? Why does it work?
- (6) Suppose now that you're not the guy behind the corner. What should you take care of if your weapon has spread? What should you therefore try to maximize?
- (7) What if the game is third person?

3.4. Engaging The Rock. Suppose that you're playing an **FPS** and there is no spread. An enemy is standing behind a small rock and can't climb on it, he can only circle around it and peek shot as shown in the next figure,

- (1) Suppose that the rock is significantly bigger than the player-dome of your enemy. To what situation is this one approximately equivalent to?
- (2) How to engage the rock then?
- (3) Suppose now and in what follows that the rock is not that big, as in figure 14. Draw one normal. Is it unique?
- (4) If the enemy is hiding behind the rock and you don't see him yet, how do you engage?
- (5) Is there any advantage to close the distance to the rock?

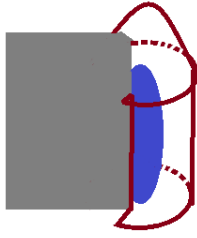


FIGURE 14. An enemy is hiding behind a rock

- (6) If your weapon has spread, and you want to engage the rock from far away, how would you proceed?
- (7) If the game is third person, is rushing straight first then taking an angle a good engagement?
- (8) Suppose now that the enemy is peeking on the right as on figure 14, and you got surprised, what do you do?

3.5. The Opening. Suppose that you're playing an **FPS** and there is no spread. The geometry is such that there is an opening. It can be a door, the entry of a tunnel, or even a window. An enemy is standing next to the opening as in the following figure,

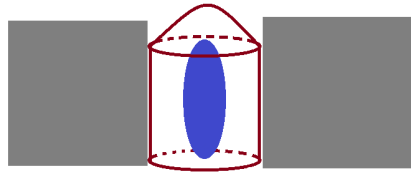


FIGURE 15. An enemy stands behind this opening

- (1) Draw a normal. Is it unique?
- (2) What are the angles from the normal that minimize the empty-dome of your enemy? Are they optimal? Is there a game-specific detail that can make them optimal?
- (3) Draw the enemy and its player-dome from an angle of 45° . Where do you position your crosshair? For this angle, what happens if the enemy pushes? If the enemy is disengaging and is going backward? At what point would you consider that the enemy has a positional advantage over you?
- (4) Suppose that there is an infinite wall on your right, what is the optimal angle?
- (5) If the enemy is camping far behind the opening on the right, explain how to kill him without pushing through the opening and while pushing through.
- (6) What if you add spread?
- (7) Suppose that your weapon is now a projectile, like a rocket launcher. Is taking an angle good in this case?

3.6. The High Ground of Low Height. Suppose that you're playing an **FPS** and there is no spread. An enemy is standing on an high ground as in figure 16, Suppose furthermore that the wall A and the floors B and C extend indefinitely such that there is no corner. The high ground is said to be of *low height* when by standing on floor C at mid to close range from wall A, you still see more than half of the model of your enemy as drawn on figure 16.

- (1) From your perspective, what plays the role of an obstacle? Same question from the perspective of the enemy.
- (2) Suppose now that the enemy is very close to the edge of the high ground. Why can you assume that the enemy can't fall? Draw the player-dome of the enemy in this situation and a normal. Is the latter unique?
- (3) Is taking an angle with respect to the normal good here? Why?

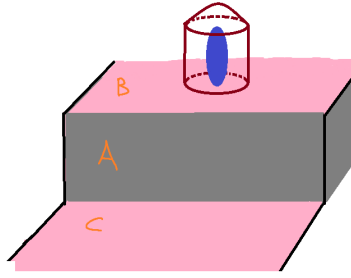


FIGURE 16. An enemy is standing on an high ground of low height

- (4) What happens if you play on the distance to the wall from your perspective and from the perspective of the enemy? Draw the 9 cases (close range, mid range, long range from the obstacle) and conclude.
- (5) What is the optimal strategy then?
- (6) What happens if there is spread?
- (7) What if the game is third person?

3.7. The High Ground of High Height. Suppose that you're playing a **FPS** and there is no spread. An enemy is standing on an high ground as in figure 17,

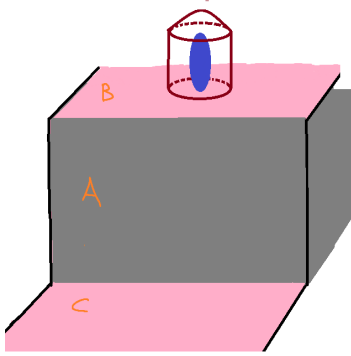


FIGURE 17. An enemy is standing on a high ground of high height

- (1) What happens if you play on the distance to the wall from your perspective and from the perspective of the enemy? Draw the 9 cases (close range, mid range, long range from the obstacle) and compare them with the High Ground of Low Height.
- (2) What is the optimal strategy then?
- (3) What should you take care of if you and the enemy are close to your respective obstacle and the game has head-shot multiplier?

3.8. The High Ground With Corner. Suppose that you're playing a **FPS** and there is no spread. An enemy is standing on an high ground with corner, as in figure 18,

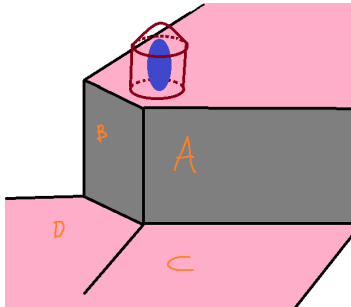


FIGURE 18. An enemy is standing on a high ground of high height

- (1) If the enemy stands closer to the edge of A (respectively of B), to what situation is this equivalent?
- (2) Suppose now that the enemy stands close to the corner joining the edge of A and B. How many normal are there? Draw them.
- (3) Should the enemy stand close to the corner to defend himself?
- (4) How are you handling his defense?

3.9. **The Slope.** Suppose that you're playing a **FPS** and there is no spread. An enemy is standing behind a slope as in figure 19,

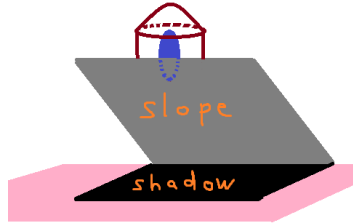


FIGURE 19. An enemy is standing on a slope and hiding half of his model-dome

- (1) Draw the normal. Is it unique?
- (2) Suppose that you're positioned far from the slope on the ground and at 0° from the normal. Does the mirror-symmetry holds true in this case? Who has the advantage in this fight?
- (3) Can you find a distance from the slope at which the mirror-symmetry holds true? What if the slope is one side of the roof of a house? Conclude.
- (4) Suppose now that you're positioned at 180° with respect to the normal. What can you say about the empty-dome and the model-dome of your enemy for this angle?
- (5) Is the previous angle optimal?
- (6) Give some optimal angles depending on the distance to the slope, and explain why they are so.

3.10. **Symmetrization.** Notice that all of the previous exercises emphasized one point of view. This exercises is not really one:

- (1) Symmetrize the previous exercises, as in, think about them from the point of view of the enemy. Try to see how to maximize damage output and how to minimize damage taken in every situations.
- (2) What if you mix different scenarios? Say, an enemy on a high ground of high height behind an opening while you're behind a rock? Think about all of these combinations.

4. SOLUTION TO THE EXERCISES

4.1. Solution to The Infinite Wall.

- (1) The normal is indeed unique, see next figure.

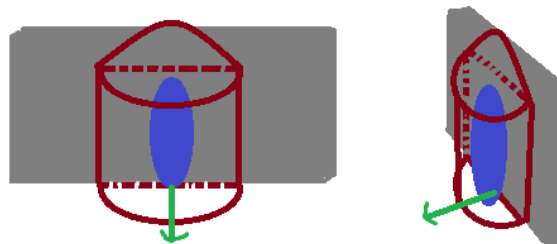


FIGURE 20. The normal is in green, and it is unique

- (2) Maximizing the model-dome here is equivalent to closing the distance because there is no obstacle. Recall that the heuristic does not work well at close range (where dodging skills are what matters), hence minimizing the empty-dome of the enemy is what is important.
- (3) $+90^\circ$ and -90° .
- (4) You wouldn't get any advantage, because you would touch the wall too and the situation would be perfectly symmetric: your self-dome would be the same as the player-dome of your enemy.
- (5) say, $+65^\circ$ or -65° . You don't want to be too close to the wall to maximize your self-dome, but you still want to be very far from the normal to increase your advantage as much as you can (and in particular, above 45°).
- (6) 45° is not optimal because at this exact value, the enemy has 5 *free directions*²³ that are not intersecting the wall. Of course, the same is true for 0° and 90° . For any other angle, like 65° , the enemy has 4 directions (instead of 3) directed toward the wall, and thus only 4 free directions. This means that if your enemy presses the 4 non-free movement keys for too long, he is going to hit the wall, which reduces his ground speed, hence making him easier to hit.

4.2. Solution to The Corner Case.

- (1) The normal is unique, as shown on the next figure,

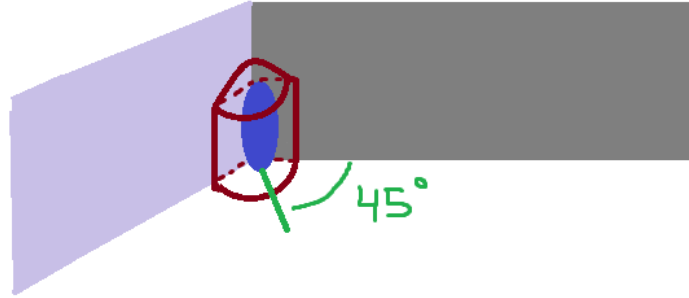


FIGURE 21. The normal is the angle bisector of the corner

- (2) $+45^\circ$ and -45° .
- (3) No, because you're making the situation symmetric because of the mirror-symmetry (for short time-scales at least).
- (4) We can think of $+20^\circ$ or -20° . Those angles are optimal because they reduce the enemy player-dome a bit and only give 2 free directions.
- (5) In this very case, the normal direction is indeed a good angle. It gives 3 free directions (which is more than 2) and maximize the player-dome, but the 3 directions are the two forward diagonals + the forward one, which are easier to hit because the relative speed difference is slower.
- (6) This isn't a good angle because it is very close to one wall so your self-dome might already be reduced. Moreover, the enemy can easily defend himself in this case because he doesn't need to move a lot to make the situation symmetric.

4.3. Solution to Behind The Corner.

- (1) No, it isn't. Note that the positive angles with respect to the normal are the ones in the anti-clockwise direction. Hence, in this case, the horizontal direction going to the right is at an angle of -90° with respect to the normal drawn.

²³a free direction is one of the eight natural directions (forward, backward, left, right, and the four diagonals) that does not intersect an obstacle if you press it

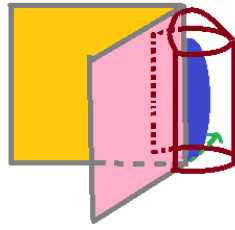


FIGURE 22. One normal

- (2) An angle of -180° (hugging the wall to the left) is bad because it minimizes your player-dome and the enemy can peek at any moment. If you're walking (and hence make no noise), it might work but it's very dangerous. Trying to reduce the angle with respect to the normal up to -90° is not bad, but by doing so you're turning the situation into two infinite walls: if the enemy is close enough to the corner and sees you, he can rotate around the corner very quickly and face you, making the situation completely symmetric by mirror-symmetry, or simply takes some distance from the wall he touches. A good angle is probably something like $(-90 - 45 = -135^\circ)$. Your crosshair must be positioned just around the corner as you move, obviously.
- (3) In this case, you're trying to maximize your (apparent surface of your) empty-dome and to minimize the one of the enemy. You can't really act on your model-dome and on the one of the enemy: after all, he is the one deciding how and when to peek. Note that when the enemy decides to peek, he will most likely see all of your empty-dome. Indeed, the perspective is such that by moving around an obstacle nearby, a lot of surface behind it is revealed. When you're far from an obstacle, because of perspective again, very few surface behind the surface is revealed by moving around it, see figure 23. The same is true for the corner, the latter being an obstacle. You therefore have much more control about what you want to see the further you are from the corner. You can therefore minimize the empty-dome of your enemy much easier than he can minimize your by increasing the distance from the corner.

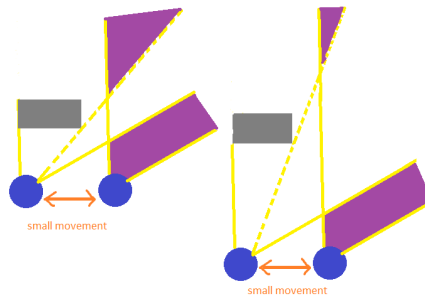


FIGURE 23. The same small movement around an obstacle (in grey) reveals more surface (in purple) if you're closer to the obstacle. Being far from the obstacle (right side) allows a better control of this revealed surface

- (4) As he peeks, because of perspective, he will have hard time controlling his model-dome and might show a lot of it. Moreover, you're the one pre-shooting the correct angle and can move a lot while your enemy is hiding behind the corner between each of his shots. You therefore have to aim much less than he does, and should play passively: try to drag the enemy around the corner and force him to be the one peeking. If you panic and shot before he peeks, you're wasting ammunition and are forced to reload before him: that's very bad. Stay calm, and remember that you have more space and the correct pre-shooting angle, so you're the one having an advantage.
- (5) Increase your distance from the corner before showing yourself: it will give you much more control over the surface revealed, and will surprise the enemy too. Even if the

enemy is very far from the corner, if you manage to increase your distance from it more than his, then thanks to perspective, you're now the one with an advantage. If you have no time to do that, you can always hug the wall and enforce an Infinite Wall situation.

- (6) If you're too far from the corner, all of your bullets are going to miss the model of your enemy. You have to increase the model-dome, and should therefore try both to come closer to the corner and to reduce the angle to the normal. This reduces your position advantage, so be cautious.
- (7) In a Third Person Shooter, the guy behind a corner has visibility, so he can peek shots much more accurately which compensates your advantage. You want to reduce the angle as much as you can to avoid being peek-killed. Moreover, some Third Person Shooters are biased toward a direction (either to the left or to the right), which makes one side easier to peek than the other (because you're forced to turn the camera more on one side to see as much as the other, which makes peeking harder).

4.4. Solution to Engaging The Rock.

- (1) If the rock is significantly bigger than the player-dome, then it takes quite some time to peek from one side to the other. The situation is consequently equivalent to two Behind The Corner situations.
- (2) Applying Behind The Corner, choosing one side (say the right side) and taking an angle on it (on the right too, you want to increase the distance from the corner) work: you're protected from the other side because of the size of the rock.
- (3) No, there is more than one normal. See next figure for one of them,

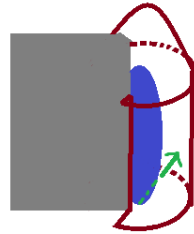


FIGURE 24. One choice of normal

- (4) You can engage him like Behind The Corner situation but with a smaller angle (in absolute value) to the normal. The purpose of the smaller angle is to forbid the enemy to peek from the other side. It also increases the distance from the rock significantly. Furthermore, because you forbid him to peek from the other side, you can position your crosshair on the “good” side of the rock. You will therefore have more space and a better expectation than your enemy if he peeks. You can also rush the rock straight: this closes the distance quickly. You should then take an angle as before. This solution is dangerous because if you engage from too far, you won't have time to protect yourself by using the rock if the enemy peeks.
- (5) By closing the distance to the rock, and by invoking the property of surface revealed by perspective as in figure 23, you actually allow yourself to hide your model-dome quicker in case of an enemy peek. Ideally, you want to close some of the distance but still stay further from the rock than the enemy is in order to maximize your empty-dome, and to allow yourself to hide if needed. Do not hesitate to “bait” the enemy by circling the rock but always keep more distance from it than your enemy: more distance means that you're slower to circle it. As a result, the enemy is still able to peek (he most likely will), and because you're the one pre-shooting the good angle and have more space, you'll hit him more than he does.

- (6) You're forced to close some of the distance in this case. Depending on the spread of your weapon, rushing straight and taking an angle then can be better than engaging from too far on one side.
- (7) Definitely not if you engage from too far. Third person allows to see behind the rock easily, and you can't predict which side the enemy is going to peek. Good timing can of course allow you to do it, but generally, you want to avoid that.
- (8) If you're close enough from the rock because you first tried to close the distance, you should try to move forward to the left to protect yourself first. Once you're protected you can start taking an angle, take more distance, and bait the peek as before. If you engaged from far on the right (the same side as the peek) you should try to reduce the angle even more, hence, going even more right to increase the model-dome of your enemy as much as you can. You might be fucked though.

4.5. Solution to the Opening.

- (1) Strictly speaking, the normal is not unique but it is legitimate to ignore the one at 180° from the one drawn on figure 25,

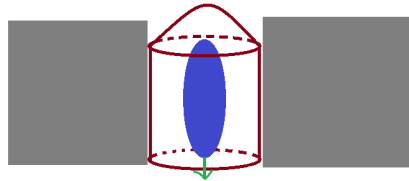


FIGURE 25. The only interesting normal

- (2) As usual, $+90^\circ$ and -90° . They are of course not optimal because you're next to a wall, and you probably can't even see the enemy from there. In Third Person Shooter, these angles become optimal because you minimize your player-dome and are still able to see the player-dome of the enemy, even if you can't hit him yet.
- (3) If you position yourself at 45° , you can aim at the very middle of the opening (the geometric point on your screen), see figure 26. By doing so, if the enemy is pushing, you don't even need to aim at all and you'll have free hits. Increasing the angle as he pushes is not bad even if you eventually get stuck on the wall, because you get more free hits. If the enemy is disengaging, you want to reduce the angle but you have to be cautious: you can get bait. Here, the two walls around the opening are playing the role of obstacles. At relatively close range, it is easy for your disengaging enemy to play around them in order to maximize your model-dome and minimize his. That is, do not follow him dumbly, maybe start increasing or decreasing the distance from the opening first to surprise him and fuck with his pre-shooting. As often, the guy who is the further from the obstacle has a positional advantage because he can manage the revealed surface by moving (and thus the enemy and his player-dome apparent surface) in an easier way than his opponent. If your opponent is further from the opening than you are, you should consider being in the worse position²⁴.
- (4) The optimal angle is -45° , that is the same position but on the left. If you really want to play on the right, then you have to keep some distance from the wall (by reducing the angle or by pushing). Otherwise, the enemy is automatically taking an angle on you in an Infinite Wall situation, and has some of his model covered by the walls of the opening. His model-dome is therefore smaller than your, which gives him an advantage (especially if the weapons have spread).
- (5) If he is camping far on the right, then the best is to start on the right too and to increase the distance as much as you can in order to invert the positional advantage. That is, once your distance from the opening is bigger than his, you can start taking an angle on the left to see him and hit him. The only way to push here is to push

²⁴unless, of course, you're camping and can surprise him, but that's another story

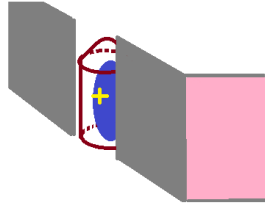


FIGURE 26. This is what you see from an angle of 45° from the normal. The crosshair is represented in yellow.

without showing yourself to the right wall of the opening, and then try to enter as fast as you can in the opening with some rectangle move.

- (6) Nothing different than usual, that is, you're forced to increase the apparent size of the model-dome which is done by closing the distance. Again, you can't afford to close it too much otherwise you would lose the positional advantage.
- (7) Projectiles have a rather big hitbox in modern games (Overwatch, Quake Champion, Fortnite, ...), and you can actually hit people with their side (imagine a rectangular shaped rocket). By taking an angle, your projectile is covering more (apparent) space of the opening for more time, which means that it is much harder to dodge. In particular, if the enemy hitbox is big too, by taking an angle of 45° and by aiming at the very center of the opening, you might cover all of it: pushing without getting hit becomes impossible. In this case, taking an angle with a projectile is even stronger than with hitscan weapons.

4.6. Solution to The High Ground of Low Height.

- (1) The wall A is playing the role of an obstacle from your perspective, while from the enemy's one it is the ground B.
- (2) You can assume that because any good player knows that it is very easy to get hit while falling, so it's better to avoid it. As a result, the player-dome of the enemy is cut by the edge of the high ground, see figure 27. The normal is unique.

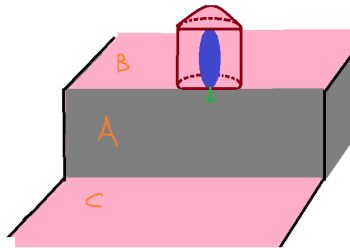


FIGURE 27. The edge of the high ground plays the role of a solid obstacle for the enemy

- (3) Yes, because you're reducing the empty-dome of the enemy while preserving a significant part of his model-dome. The bigger the angle, the smaller the empty-dome of your enemy, which means that it is eventually enough to aim at a geometric point (as in The Opening situation) to obtain free hits. Obviously, if your angle is too big (say, 90°) you're reducing your empty-dome too, so it isn't clear (yet) if large angles are good.
- (4) By looking at figure 28 it is easy to notice that:
 - the sight of your enemy increases as he gets closer to the edge of B,
 - your sight increases as you go further from the wall A,
 - when both of you are at mid distance from your respective obstacle, your (apparent) model-dome is bigger than the one of your enemy. You're therefore at a disadvantage if you don't take an angle on him,

- the enemy can reduce his model-dome significantly without reducing your (apparent one) by increasing the distance from the edge of the high ground,
- even though getting closer to the wall reduces your eye-sight, it reduces your (apparent) model-dome more than the one of your enemy,

Closing the distance to the wall is good even without taking a too big angle, because the difference of (apparent) size of model-dome favors you anyway.

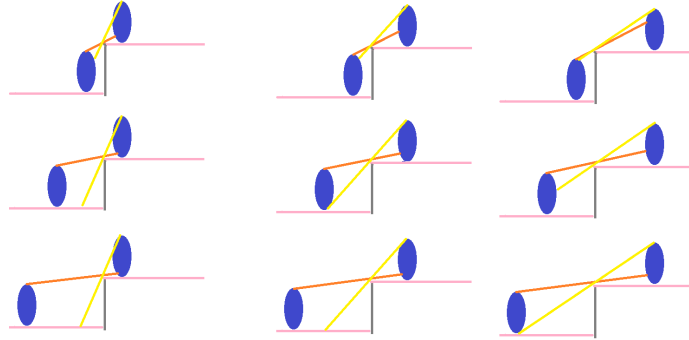


FIGURE 28. Perspective for close, mid, and long range. The orange line is the boundary of your sight, while the yellow line is the boundary of the sight of your enemy

- (5) The optimal strategy is to take an angle and to try to close the distance from A. By doing so, if your enemy still takes the fight, he has to close the distance to the edge. There, his (apparent) model-dome is bigger than your, and thanks to the angle, it is enough for you to aim at a geometric point to hit a lot. Moreover, by coming close to the edge, your opponent can fall, which favors you even more. Now, the enemy can defend himself by increasing his distance from the edge (hence baiting you to +back from the wall too) or by forbidding you to take an angle (by mirroring your global lateral movement). Be cautious of this kind of bait, at long and mid range, your model-dome is bigger than his, so your enemy needs less precision to aim at you and even though you have more space. Also, if the enemy is playing perfectly with the distance to the edge, he can reduce his model-dome to the size of his head, which makes your aim very hard. Notice that this is our first example (in a **FPS**) where the best position is the one closer to the obstacle. This is actually quite general with low-ground positions. This also means that you should work on your (large) vertical angle: being better at them than your opponent is already a big advantage, so if you add a better positioning, it increases your odds of winning the fight by a huge margin.
- (6) If there is spread, you should position yourself so that the model-dome of your enemy is big enough for your weapon to hit. Aiming at the very center of a small model is very hard, so keep that in mind. Playing long range can work if the spread is not too big because you're even with your enemy on the size of your respective model-dome. Playing close-range by taking a big angle is better but harder.
- (7) If the game is third person, the visibility of your enemy is much better than your. In particular, he can play "perfectly" (as described above) in a much easier way than if the game was first person. As a result, even if you take an angle, playing close to mid range is bad. Playing long range evens visibility out, so you can always try that (and remember to take an angle).

4.7. Solution to The High Ground of High Height.

- (1) as you can see on figure 29, the situation is very similar to the one of the high ground of low height. The difference being that you need to be closer to the wall to gain an advantage, and even further from it to evens out the apparent model-domes. That is, the situation is the same but everything is more "extrem".

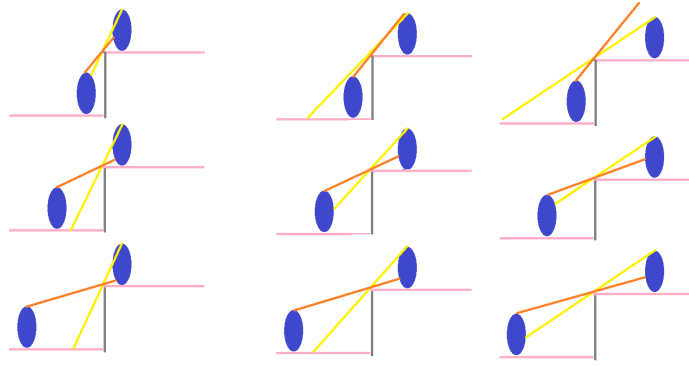


FIGURE 29. Perspective for a high ground of high height

- (2) The same as before, but you need to keep in mind that you have to come even closer to the wall, or take even more distance if you want to fight at long range.
- (3) When you're both close, the enemy (high ground position) only sees your head. If the head-shot multiplier is 2, this means that you have to hit at least twice as him. You therefore need good intuition about how easy it is for you to out damage your opponent in this situation.

4.8. Solution of The High Ground With Corner.

- (1) This is clearly equivalent to The High Ground situation with a normal orthogonal to the edge of A (respectively to the edge of B). As in, you can completely ignore the corner in this case.
- (2) If the enemy stands exactly on the corner, then there are two normals: the ones coming from The High Ground situation with edge A and B.
- (3) An enemy should never stand exactly on the corner, because it is the place that constraints his player-dome the most. Still, he should indeed try to play around the corner and the edge of A and B, to gain more space and to forbid you to take an angle on him.
- (4) You should imagine that the normals are updating in real time as the enemy swaps position from the edge of A to the edge of B. That is, supposing that your enemy started away from the corner and closer to the edge of A, you should take an angle on the zone D. As your enemy defends himself and swaps position near the edge of B, you should swap too: it is now time to take an angle from the zone C. Obviously, the closer you stand from the walls, the faster you are at changing zones, so try to play this situation close range. If you play this situation at mid or long range, then it is very easy for your opponent to deny your angle, making the situation an aim duel.

4.9. Solution to The slope.

- (1) The normal is unique as shown on figure 30,

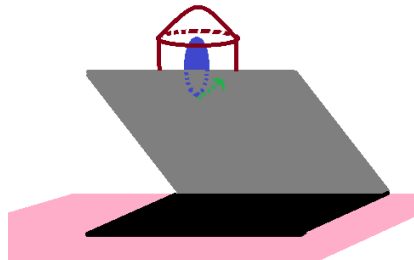


FIGURE 30. The normal is unique and points in the direction you're looking (it is therefore behind the enemy)

- (2) No, because if the enemy moves forward or backward, his pitch changes considerably. This means that the height of (the apparent surface of) the model-dome of your enemy is bigger than your own. Moreover, the enemy stands on the middle of the slope, hence the width of his player-dome is as big as your. In this case, you're at a disadvantage.
- (3) If the slope touches the ground then no. If the slope is one side of a roof of a house, then yes, as shown on figure 31, At this very precise distance, your empty-dome is

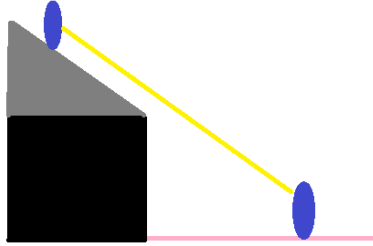


FIGURE 31. At this distance from the roof the mirror-symmetry holds true from your perspective

bigger vertically than the one of the enemy, and you consequently have the better position.

- (4) The empty-dome is not minimized, the model-dome might be maximized from this side, but it is rarely enough to compensate for the (apparent) space you give to your enemy. Furthermore, there is a left-right symmetry of the player-dome, which means that you cannot predict anything about the horizontal motion of your enemy. This makes your aim harder than it can be.
- (5) Surely not.
- (6) At mid and long range, one can think of $\pm 135^\circ$. At this angle, you minimize the empty-dome of your enemy, but the model-dome is quite small too so it might be hard land your shots, especially if there is spread. A smaller angle (in absolute value) but bigger than 90° will increase the model-dome of your enemy but also the empty-dome. Do some test in real game to gain some intuition about what is best for your aiming skills. In particular, the position you find optimal depends of the slope of the slope²⁵. At close range, because the slope also plays the role of a high ground, closing the distance from the side will obviously give you an advantage. From this position, the model-dome of your enemy will be bigger than your (because your lower body is now hidden by the slope, while his can't).

²⁵pun intended